

URL Shortener service – Complete Testing Guide

Table of Contents

1. Pre-requisites
2. Setup
3. Feature Testing (12 Features)
4. Automated Tests
5. Troubleshooting

1. Pre-requisites

Requirement	Version	How to Check
Python	3.11 or higher	python --version
pip	Any (comes with python)	pip --version

Note:

- Windows: Download Python from <https://www.python.org/> -> During install, check “Add to Path”
- Mac: Open Terminal -> brew install python3 or download from <https://www.python.org/>

Command Differences:

OS	Python Command	Pip command
Windows	python	pip
Mac	Python3	Pip3

```
hp@LAPTOP-3JLRI300 MINGW64 ~  
• $ python --version  
Python 3.13.14  
  
hp@LAPTOP-3JLRI300 MINGW64 ~  
• $ pip --version  
pip 26.1.2 from C:\Program Files\WindowsApps\PythonSoftwareFoundation.Python.3.13_3.13.3824.0_x64__qbz5n2kfra8p0\Lib\site-packages  
\pip (python 3.13)  
.
```

2. Setup

Step 1: Download the Project folder and Open Terminal (Command Prompt or VS Code) and go to Project folder and install Dependencies, start the server, open the app in browser.

cd url-shortener-service

pip install -r requirements.txt

```
hp@LAPTOP-3JLRI300 MINGW64 ~/Downloads/hemanth/url-shortener-service  
• $ ll  
total 15  
drwxr-xr-x 1 hp 197121  0 Jul 30 05:50 app/  
drwxr-xr-x 1 hp 197121  0 Jul 30 05:50 docs/  
-rw-r--r-- 1 hp 197121 550 Jul 30 05:50 main.py  
-rw-r--r-- 1 hp 197121 8052 Jul 30 05:50 README.md  
-rw-r--r-- 1 hp 197121  38 Jul 30 05:50 requirements.txt  
-rw-r--r-- 1 hp 197121 162 Jul 30 05:50 START_HERE.txt  
drwxr-xr-x 1 hp 197121  0 Jul 30 05:50 tests/  
  
hp@LAPTOP-3JLRI300 MINGW64 ~/Downloads/hemanth/url-shortener-service  
• $ pip install -r requirements.txt
```

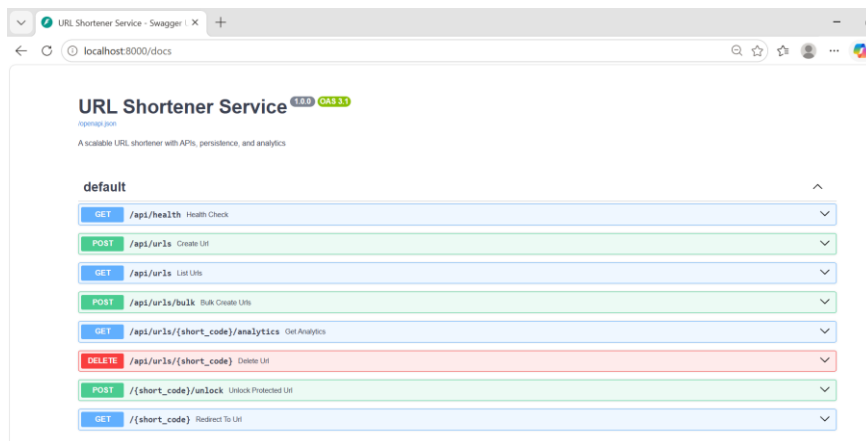
Start the server:

python main.py

```
hp@LAPTOP-3JLRI300 MINGW64 ~/Downloads/hemanth/url-shortener-service
$ python main.py
INFO: Will watch for changes in these directories: ['C:\\Users\\hp\\Downloads\\hemanth\\url-shortener-service']
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Started reloader process [10356] using StatReload
INFO: Started server process [10460]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

Open the App in browser:

Go to: <http://localhost:8000/docs>



Step-3: Feature Testing

Feature 1: Health Check

**** Business Case:**** verify the server is running

**** Steps:****

1. In UI, click `GET /api/health`
2. Click "Try it out"
3. Click "Execute"

**** Expected Output:**** "status": "healthy"

Request URL	
<code>http://localhost:8000/api/health</code>	
Server response	
Code	Details
200	Response body<pre>{ "status": "healthy"}</pre>

Feature 2: Create Short URL

**** Business Case:**** Converts a long website link into a short, easy-to-share link.

****Steps:****

1. Click `POST /api/urls`
2. Click "Try it out"
3. Clear the text box, paste:

```
{
  "url": "https://www.schwab.com",
  "custom_code": "schwab"
}
```

4. Click "Execute"

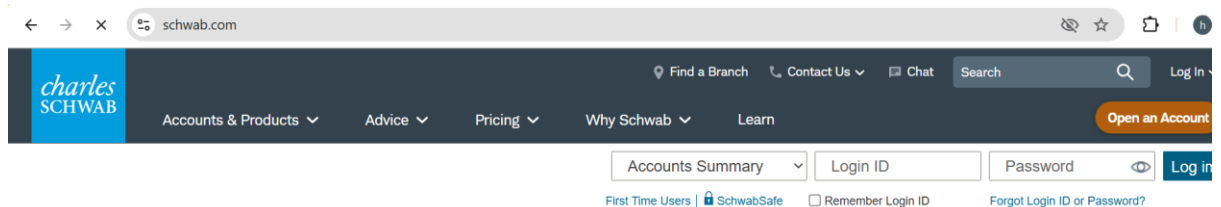
**** Expected Output:****

Server response	
Code	Details
201	Response body <pre>{ "short_code": "schwab", "short_url": "http://localhost:8000/schwab", "original_url": "https://www.schwab.com", "created_at": "2026-07-30 02:19:08", "max_clicks": null, "expires_at": null, "is_protected": false }</pre>

5. Test the short URL (Customer Clicks short link -> arrives at real website).

In browser, type: <http://localhost:8000/schwab>

Browser redirects to Schwab official Home Page.



Feature 3: List all URLs

**** Business Case:**** Admin Dashboard showing all active links.

****Steps:****

1. Click `POST /api/urls`
2. Click "Try it out"

Code	Details
200	Response body <pre>{ "urls": [{ "short_code": "schwab", "short_url": "http://localhost:8000/schwab", "original_url": "https://www.schwab.com", "created_at": "2026-07-30 02:19:08", "max_clicks": null, "expires_at": null, "is_protected": false }] }</pre>

Feature 4: Bulk Create URLs

****Business Case:**** Company might have 100+ product pages. Creating them one by one takes hours. Bulk create does it in one click. (Short url codes will be auto generated)

****steps:****

1. Click `POST /api/urls/bulk`
2. Click "Try it out"
3. Clear the text box, paste:

```
{
  "urls": [
    "https://www.schwab.com/trading",
    "https://www.schwab.com/retirement"
  ]
}
```

POST /api/urls/bulk Bulk Create Urls

Create multiple short URLs at once (max 50).

Parameters

No parameters

Request body required

Edit Value | Schema

```
{
  "urls": [
    "https://www.schwab.com/trading",
    "https://www.schwab.com/retirement"
  ]
}
```

4. Click "Execute"

Server response

Code	Details
201	Response body <pre>{ "created": 2, "results": [{ "short_code": "AJ5nHFq", "short_url": "http://localhost:8000/AJ5nHFq", "original_url": "https://www.schwab.com/trading", "created_at": "2026-07-30 03:11:52", "max_clicks": null, "expires_at": null, "is_protected": false }, { "short_code": "GSyPl3H", "short_url": "http://localhost:8000/GSyPl3H", "original_url": "https://www.schwab.com/retirement", "created_at": "2026-07-30 03:11:52", "max_clicks": null, "expires_at": null, "is_protected": false }] }</pre>

5. Multiple Short URLs will be created. We can test in browser with short URL. It will redirect to original Page.

<http://localhost:8000/AJ5nHFq> -> <https://www.schwab.com/trading>

<http://localhost:8000/GSyPl3H> -> <https://www.schwab.com/retirement>

Feature 5: Analytics

**** Business Case:**** It helps in understanding customer engagement – how many people are interested and interacting with the shared links.

****Steps:****

1. Click `POST /api/urls/{short\_code}/analytics`
2. Click "Try it out"
3. Type short url word "schwab" in the short_code field
4. Click "Execute"

The screenshot shows a REST client interface for the endpoint `GET /api/urls/{short_code}/analytics`. The parameters section shows `short_code` (string, path) with the value `schwab`. The response status is 200, and the response body is a JSON object:

```
{
  "short_code": "schwab",
  "original_url": "https://www.schwab.com",
  "total_clicks": 3,
  "max_clicks": null,
  "clicks_remaining": null,
  "expires_at": null,
  "is_expired": false,
  "created_at": "2026-07-30 02:19:08",
  "clicks_by_date": [
    {
      "date": "2026-07-30",
      "count": 3
    }
  ],
  "recent_clicks": [
    {
      "clicked_at": "2026-07-30 02:28:26",
      "ip": "127.0.0.1",
      "user_agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36",
      "referrer": null
    }
  ]
}
```

We can see how many people clicked our product link, at what time they clicked and from which location. It helps in understanding customer engagement – how many people are interested and interacting with the shared links.

Feature 6: Delete URL

**** Business Case:**** Removes a link that no longer needed.

****Steps:****

1. Click `POST /api/urls/{short\_code}`
2. Click "Try it out"
3. Type short url word "schwab" in the short_code field
4. Click "Execute"

localhost:8000/docs#/default/delete_url_api_urls_short_code_delete

DELETE /api/urls/{short_code} Delete Url

Delete a short URL (soft delete).

Parameters

Name	Description
short_code * required string (path)	schwab

Execute

Expected: Code 204 (no body means short URL deleted)

Verify: Open <http://localhost:8000/schwab> in browser - should show "not found" now.

← ↻ ⓘ localhost:8000/schwab

Pretty-print ☐

```
{"detail": "Short URL not found, expired, or click limit reached"}
```

Feature 7: Unlock Protected URL – Password Protection

**** Business Case:**** This is necessary to share a confidential link like a salary document – only people with the password can access the confidential document.

****Steps:****

- Create a password protected URL:
Go to POST /api/urls -> Try it out -> Paste:


```
{
  "url": "https://www.schwab.com/wealth-management",
  "custom_code": "wealth-advisor",
  "password": "schwab123"
}
```

POST /api/urls Create Url

Create a new short URL. Options: max_clicks, expires_in_hours, password.

Parameters

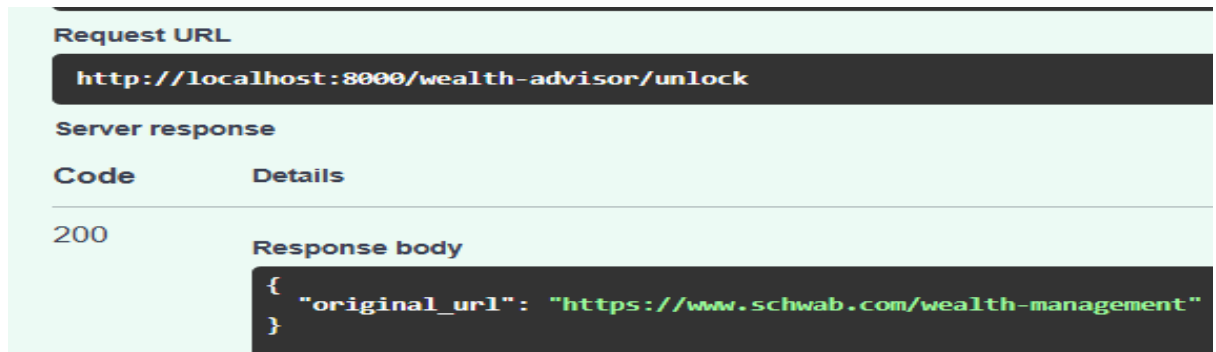
No parameters

Request body required

Edit Value | Schema

```
{
  "url": "https://www.schwab.com/wealth-management",
  "custom_code": "wealth-advisor",
  "password": "schwab123"
}
```

2. Try – Unlocking:
Go to POST /{short_code}/unlock – Try it out: Enter the short code and password.
Short-code: wealth-advisor
Passwords: schwab123



Provides a secure way to access password-protected links – verifies the password and reveals the original URL if correct.

Feature 8: Rate Limiting (Anti-Spam protection)

**** Business Case:**** If one person keeps creating too many links too fast, the system will block them for 1 minute. This keeps the system safe and fair for everyone.

We have configured In a way 10 requests per minute.

****Steps:****

1. Go to POST /api/urls -> Try it out -> Paste:

```
{
  "url": https://www.schwab.com/
}
```
2. Click Execute – you get 201 (success)
3. Keep clicking after 10 times – you will get (Rate limit exceeded. Max 10 requests per minute)



This can be used to Automatically blocks users who make too many requests – Protects the system from spam, bots.

4. Automated Tests:

To check if all features are working correctly

****Steps:****

1. Open a new terminal
2. Go to project folder
3. Run: **python -m pytest tests/ -v**

Expected: all test cases passed (all green)

```
hp@LAPTOP-3JLRI300 MINGW64 ~/Downloads/hemanth/url-shortener-service
$ python -m pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.13.14, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\hp\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.13_
qbz5n2kfra8p0\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\hp\Downloads\hemanth\url-shortener-service
plugins: anyio-4.14.2
collected 27 items

tests/test_api.py::test_create_short_url PASSED [ 3%]
tests/test_api.py::test_create_with_custom_code PASSED [ 7%]
tests/test_api.py::test_create_invalid_url PASSED [ 11%]
tests/test_api.py::test_create_duplicate_code PASSED [ 14%]
tests/test_api.py::test_redirect PASSED [ 18%]
tests/test_api.py::test_redirect_not_found PASSED [ 22%]
tests/test_api.py::test_analytics PASSED [ 25%]
tests/test_api.py::test_analytics_not_found PASSED [ 29%]
tests/test_api.py::test_list_urls PASSED [ 33%]
tests/test_api.py::test_delete_url PASSED [ 37%]
tests/test_api.py::test_delete_not_found PASSED [ 40%]
tests/test_api.py::test_health PASSED [ 44%]
tests/test_api.py::test_create_with_click_limit PASSED [ 48%]
tests/test_api.py::test_click_limit_works PASSED [ 51%]
tests/test_api.py::test_click_limit_shows_in_analytics PASSED [ 55%]
tests/test_api.py::test_no_click_limit_works_forever PASSED [ 59%]
tests/test_api.py::test_create_with_expiration PASSED [ 62%]
tests/test_api.py::test_non_expired_url_works PASSED [ 66%]
tests/test_api.py::test_create_with_password PASSED [ 70%]
tests/test_api.py::test_password_required_on_access PASSED [ 74%]
tests/test_api.py::test_wrong_password_rejected PASSED [ 77%]
tests/test_api.py::test_correct_password_works PASSED [ 81%]
tests/test_api.py::test_unlock_endpoint PASSED [ 85%]
tests/test_api.py::test_bulk_create PASSED [ 88%]
tests/test_api.py::test_bulk_create_with_expiry PASSED [ 92%]
tests/test_api.py::test_blocked_domain_rejected PASSED [ 96%]
tests/test_api.py::test_safe_domain_allowed PASSED [100%]
```

5. Troubleshooting:

Problem	Solution
`python: command not found`	Try `python3` instead`
500 Internal server error	Delete `data/` folder and restart server
422 Error	Check JSON – no trailing commas after last value
Port 8000 in use	Close other apps using port 8000, or restart